SAKSHAM GUPTA

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EDUCATION

Manipal Institute of Technology, Manipal *Manipal, Karnataka • May 2027*

B.Tech** in **Information Technology (CGPA: 9.12/10.0)

Delhi Public School R.K. Puram *New Delhi • 2023*

Class 12 - CBSE (90.2%) • Class 10 - CBSE (93.5%)

TECHNICAL SKILLS

Computer Networks & Protocols: TCP/IP, OSI Model, IP Addressing & Subnetting, Routing Protocols (OSPF, BGP, RIP, VRRP), Switching (VLANs, xSTP, LACP, 802.1Q), ARP, ICMP, DNS, DHCP, Wireshark Packet Analysis

Systems & Operating Systems: Linux/UNIX Administration, OS Internals, Process & Memory Management, Socket Programming, Multi-threading & Concurrency, Bash/Shell Scripting

Network Security & Infrastructure: Firewalls, Access Control Lists (ACLs), VPNs, Cryptography (AES-256), Intrusion Detection, Docker, CI/CD, Git, AWS

Programming Languages: Python, C++, Java, C, Go, JavaScript, TypeScript, SQL

Core CS & Methodologies: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Distributed Systems, Software Testing & Debugging

ACHIEVEMENTS & CERTIFICATIONS

- **CCNA (Cisco Certified Network Associate)** - Introduction to Networks, Switching/Routing/Wireless Essentials, Enterprise Networking, Security & Automation (Cisco NetAcad)
- **Achievers Scholarship** - Ranked in Top 5% of Information Technology cohort at MIT Manipal
- **Academic Excellence** - Maintained consistent CGPA of 9.12/10 across all semesters
- **Open Source Contributor** - Flatcar Container Linux, Zephyr RTOS (Embedded Networking), and Zowe (Open Mainframe Project)
- **Professional Certifications** - Google Cloud Digital Leader & AWS Generative AI Applications (Coursera)

EXPERIENCE

Software Development Intern *June – July, 2025*

Indian Railway Catering and Tourism Corporation (IRCTC)

- Analyzed an enterprise ticketing infrastructure handling 2M+ daily transactions, mapping network data flow pipelines and identifying distributed latency bottlenecks
- Evaluated microservices communication and distributed tracing protocols across network segments, executing automated load tests using Python to simulate concurrent traffic
- Optimized relational database queries by 25% via execution plan profiling and index tuning; documented network monitoring and system observability insights

Software Intern *June – July, 2026*

AeroSportsParks

- Designed and implemented a Redis-based caching layer for recurring sessions, slashing network read latency and preventing redundant database queries
- Built an automated schema metadata visualizer fetching live relationships from AWS EC2 instances using object-oriented design principles
- Engineered automated data migration scripts to safely transition production records across database architectures with zero packet or data loss

PROJECTS

SafePay | Network Security & Transaction Fraud Detection System | GitHub **•** View Live

Python, Flask, Cryptography (AES-256), REST APIs, MongoDB, Scikit-learn, Network Security

- Engineered an end-to-end secure transaction verification pipeline using AES-256 payload encryption, digital signatures, and secure password hashing
- Built an ML fraud detection engine scoring live incoming network payloads with sub-50ms inference latency, achieving 85%+ classification accuracy
- Exposed low-latency REST endpoints with strict input validation, preventing unauthorized request injection and securing financial data transmission

QuickDrop | Real-Time High-Throughput Network Streaming Platform | GitHub **•** View Live

Node.js, WebSockets, Socket.io, Express, Network Protocols, MongoDB, React

- Built a real-time event streaming and data transfer platform using bi-directional WebSockets, delivering sub-100ms packet delivery across concurrent clients
- Implemented socket room state management with automated session timeouts, minimizing server memory footprints under high concurrent connections
- Engineered network error recovery with automatic socket reconnections and chunked binary file transmissions with integrity verification

Typeform | Full-Stack Form Engine & Systems Architecture | GitHub **•** View Live

Next.js, TypeScript, FastAPI, SQLAlchemy, SQLite, Azure VM, Linux, Docker

- Architected and deployed a multi-tier service on an Azure Linux VM, managing network port configurations, reverse proxy routing, and persistent storage
- Engineered dynamic conditional branching logic with sub-second API round-trip times and clean client-server state persistence

LEADERSHIP & ACTIVITIES

Organizing Committee Member — Sports (Chess) *March 2025*

Revels'25, MIT Manipal

- Led tournament operations and bracket systems for 200+ participants across tiered stages, coordinating with a 12-member operations team